

ABSTRACT

This report presents the design and implementation of a desktop-based Payroll Management System developed using Java Swing, JDBC, and MySQL. The primary objective of the system is to simplify the process of managing employee payroll records by providing an efficient and user-friendly interface, thereby reducing manual effort and improving data accuracy.

The application supports essential operations such as adding, viewing, updating, and deleting employee records through a graphical user interface. Each record includes details such as employee name, department, and salary. The system is structured using a layered architecture consisting of Data Transfer Object (DTO), Data Access Object (DAO), Service, and User Interface (UI) layers. This separation of concerns ensures modularity, ease of maintenance, and scalability of the application.

The DAO layer handles all database interactions using JDBC, enabling secure and efficient communication with the MySQL database. The service layer incorporates business logic and validations to ensure correctness and consistency of payroll data. The UI layer, built using Java Swing, provides an intuitive platform for users to interact with the system and perform operations seamlessly.

This project demonstrates the integration of a Java-based graphical interface with a relational database system and highlights the importance of structured software design in developing reliable and maintainable payroll management applications.